Yuxuan Jiang

yuxuanj1@umbc.edu yuxuanjiang1.github.io yuxuanjiang1 +1-551-998-5415

Final-year Computer Science Ph.D. student specializing in **LLM post-training and recursive self-improvement**, with **3+ first-author publications at NeurIPS, ACL, and EMNLP**. Recently completed a **research internship at Emergence AI**.

Education

University of Maryland, Baltimore County, Baltimore, MD — Sep. 2023 – Present
Ph.D. in Computer Science, expected May 2027 — Advisor: Prof. Francis Ferraro

New York University, New York, NY — Jan. 2021 – Dec. 2022
M.S. in Computer Science

Beijing Language and Culture University, Beijing, China — Sep. 2016 – Jul. 2020
B.S. in Information Systems

Research Experience

Emergence AI, New York, USA — May 2026 – Aug. 2026
Research Intern

- Developed **SO-RSI**, a recursive self-improvement framework for Lean 4 theorem proving and Verus code generation that investigates why workflow edits fail, rather than repeatedly patching compiler errors.
- Implemented a lightweight detector for recurring failures, opposing edits, and unmet expectations; maintained investigation histories across RSI rounds and used diagnostic probes to test competing explanations before modifying prompts, retrieval, or repair policies.
- Constructed evaluation sets with **92/48 development/test problems for Lean 4** and **113/46 for Verus**; evaluated each method over three runs with a 24-hour optimization budget per run.
- Improved held-out **pass rates by 21.8 percentage points on Lean 4** (49.4% → 71.2%) and **25.8 points on Verus** (33.4% → 59.2%) over Naive RSI under the same 24-hour optimization budget, using evidence-guided workflow edits.

Selected Research Projects

Reusable Process Rewards for Reasoning and Tool Use *PyTorch, GRPO, LLM-as-a-Judge*

- Developed **SCRIBE** (EMNLP 2026 Main), which extracts reusable skill prototypes from reasoning trajectories and converts them into explicit subgoal-level scoring rubrics without requiring per-instance reference answers.
- Routed generated subgoals to shared rubrics and used the resulting process rewards for GRPO; cross-instance scoring agreement improved from 69.4% with instance-level rubrics to 86.8% with SCRIBE.
- Improved Qwen3-4B performance from 43.3% to 63.3% on AIME25 and from 33.0% to 51.3% on BFCL V4 overall, relative to the base model.

On-Policy Knowledge Distillation for Large Reasoning Models *PyTorch, FSDP2, SGLang, Ray*

- Developed **TOPD** (Trajectory Aware On-Policy Distillation), which regenerates student continuations after teacher-suggested corrections and allocates teacher supervision across reasoning positions using loss-aware optimal transport.
- Distilled **Qwen3-30B-A3B into Qwen3-4B** on **4× H100 (80GB)** GPUs, using 20k teacher-generated solutions for warm-start distillation and 50k on-policy prompts with four student rollouts per prompt; improved average pass@1 across AIME24, AIME25, and HMMT25-Feb from **47.8% to 57.8%** over ordinary on-policy distillation.
- Identified **"rock tokens"** (NeurIPS 2026) — tokens with persistently high distillation loss — and used gradient analysis and targeted interventions to distinguish student–teacher mismatch from useful reasoning supervision.
- Implemented TOPD within KDFlow using SGLang, FSDP2, and Ray; coordinated student rollouts, teacher inference, and student updates on shared GPUs through CPU offloading and engine sleep/wakeup, with BF16 training and gradient checkpointing.

Selected Research Projects (Continued)

Efficient Post-training for Large Reasoning Models *PyTorch, vLLM, LLaMA-Factory*

- Proposed **Distilled Reasoning Pruning (DRP)**, a step-level knowledge distillation method that decomposes chain-of-thought reasoning into mathematical skill units and prunes steps that contribute no new skill information.
- Reduced **generated tokens per response by >20%** while improving pass@1 over standard distillation baselines: **MATH500 64.0% → 70.5%** and **AIME24 26.7% → 50.0%**, improving both token efficiency and reasoning accuracy.
- Conducted ablations on step selection criteria, skill taxonomy design, and distillation temperature across 7B–30B teacher models, identifying conditions under which step pruning helps vs. hurts reasoning quality.

First-Author Publications

1. "Second-Order Problem Solving for Recursive Self-Improvement in Formal Verification". *arXiv preprint, 2026.*
2. "Bridging Reasoning Trajectories in On-Policy Distillation via Near-Future Guidance". *arXiv preprint, 2026.*
3. "Cornerstones or Stumbling Blocks? Deciphering the Rock Tokens in On-Policy Distillation". **NeurIPS 2026**.
4. "SCRIBE: Learning Reusable Skill Rubrics for Process Reward Models". **EMNLP 2026 Main Conference**.
5. "DRP: Distilled Reasoning Pruning with Skill-aware Step Decomposition for Efficient Large Reasoning Models". *ACL 2026 Findings.*
6. "Beyond Math: Stories as a Testbed for Memorization-Constrained Reasoning in LLMs". *EACL 2026* (**Oral**).

Technical Skills

- **Research:** LLM post-training, on-policy distillation, GRPO, process reward modeling, agent evaluation
- **Programming:** Python, C++, SQL, Shell
- **Training & Inference:** PyTorch, Hugging Face, PEFT, TRL, DeepSpeed, FSDP2, SGLang, vLLM
- **Infrastructure:** Ray, Slurm, Linux, Docker, Git, AWS, GCP